

RESEARCH ARTICLE

Comparing Generative AI Literature Reviews Versus Human-Led Systematic Literature Reviews: A Case Study on Big Data Research

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ABSTRACT Generative Artificial Intelligence (GenAI) and Large Language Models (LLMs) are transforming research methodologies, including Systematic Literature Reviews (SLRs). While traditional, human-led SLRs are labor-intensive, AI-driven approaches promise efficiency and scalability. However, the reliability and accuracy of AI-generated literature reviews remain uncertain. This study investigates the performance of GPT-4-powered Consensus in conducting an SLR on Big Data research, comparing its results with a manually conducted SLR. To evaluate Consensus, we analyzed its ability to detect relevant studies, extract key insights, and synthesize findings. Our human-led SLR identified 32 primary studies (PSs) and 207 related works, whereas Consensus detected 22 PSs, with 16 overlapping with the manual selection and 5 false positives. The AI-selected studies had an average citation count of 202 per study, significantly higher than the 64.4 citations per study in the manual SLR, indicating a possible bias toward highly cited papers. However, none of the 32 PSs selected manually were included in the AI-generated results, highlighting recall and selection accuracy limitations. Key findings reveal that Consensus accelerates literature retrieval but suffers from hallucinations, reference inaccuracies, and limited critical analysis. Specifically, it failed to capture nuanced research challenges and missed important application domains. Precision, recall, and F1 scores of the AI-selected studies were 76.2%, 38.1%, and 50.6%, respectively, demonstrating that while AI retrieves relevant papers with high precision, it lacks comprehensiveness. To mitigate these limitations, we propose a hybrid AI-human SLR framework, where AI enhances search efficiency while human reviewers ensure rigor and validity. While AI can support literature reviews, human oversight remains essential for ensuring accuracy and depth. Future research should assess AI-assisted SLRs across multiple disciplines to validate generalizability and explore domain-specific LLMs for improved performance.

INDEX TERMS AI-assisted research, big data, generative AI, large language models, systematic literature review.

I. INTRODUCTION

Generative Artificial Intelligence (GenAI) and Large Language Models (LLMs) are drastically changing the landscape of scientific research by automating complex tasks, starting from solving coding problems [1] or conducting Systematic Literature Reviews (SLRs). Every day, new GenAI models (like Google Bard, GPT-4, or OpenAI o1) are released, and it

is now fundamental to understand how to adopt and leverage these systems to streamline traditional, human-led tasks.

The potentiality of using GenAI in the field of scientific literature review is evident. Traditional SLRs are time-consuming and labor-intensive, involving the manual identification, screening, and synthesis of large volumes of research papers. GenAI models offer the potential to revolutionize this process by significantly accelerating review timelines, enhancing accuracy, and improving the comprehensiveness of the reviews. LLMs and GenAI can assist or

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even replace some of the manual steps in SLRs since these models are capable of processing vast amounts of information quickly, identifying relevant studies, extracting key insights, and presenting these insights in coherent summaries. This ability is particularly valuable in an era of scientific publication overload, where the number of published studies is growing exponentially across disciplines. Automating some steps in an SLR workflow can address challenges such as information overload, human error, and the inherent biases that arise in manual reviews. However, despite their efficiency, the possibility of replacing all the human-led steps when conducting an SLR is utopian right now. There is still a debate about whether LLMs can match or surpass human reviewers in terms of accuracy, critical analysis, and the interpretation of nuanced, context-dependent information.

GenAI and LLMs show significant promise in enhancing SLRs by automating literature searches and screening processes. AI-powered tools are increasingly being used to streamline Systematic Literature Reviews (SLRs), leveraging Large Language Models (LLMs) to automate study selection, data extraction, and synthesis. Recent research highlights both the potential benefits and limitations of these AI-assisted approaches.

Several studies explore the role of domain-specific AI models in improving literature reviews. Susnjak proposes fine-tuned LLMs specifically trained for research synthesis, demonstrating how AI can enhance efficiency while minimizing hallucinations and reference inaccuracies [3], [4]. Similarly, Bolanos et al. analyze existing SLR automation tools, identifying trends and key challenges in AI-based literature reviews [5].

AI also improves systematic review efficiency. Tran et al. report a 97.1% sensitivity rate for AI-driven abstract screening, suggesting that AI can significantly reduce manual workload [6]. Castillo-Segura et al. further highlight how Generative AI aids in classifying and summarizing studies, but caution against its lack of critical analysis and overreliance on citation metrics [7].

Despite these advancements, AI-assisted SLRs face critical challenges. Studies reveal that LLMs often favor highly cited papers, potentially neglecting emerging research. Moreover, AI lacks contextual understanding and domain-specific judgment, making human oversight essential [8]. Iannantuono et al. advocate for a hybrid approach, where AI supports literature retrieval, but final validation and synthesis remain human-led [9].

To address these challenges, researchers emphasize the need for better AI-human collaboration frameworks and domain-specific LLMs, and the use of AI tools that can streamline the systematic review process by allowing authors rapid access to relevant manuscripts, thereby enhancing the efficiency of comprehensive literature reviews [11]. Future work should explore cross-disciplinary validation, assessing AI-assisted SLRs in diverse research fields, from healthcare and finance to engineering and social sciences [10], [13] and

the risk of bias analysis from a randomized controlled trial when using ChatGPT in meta-analyses [12].

This paper builds upon these findings by empirically evaluating GPT-4-powered Consensus in conducting an SLR on Big Data research, comparing AI-generated results with human-led reviews to assess accuracy, efficiency, and limitations.

This paper practically explores these issues by investigating the effectiveness of the Consensus model [https://consensus.app], a free fine-tuned version of GPT-4 specialized for the world's scientific literature, in conducting SLRs, with a focus on its efficiency, accuracy, and depth of analysis compared to traditional human-led reviews. To this end, we show the results of our SLR [2] if compared to the outputs generated by Consensus from a quantitative and a qualitative point-of-view. We focus on the 3 research questions (RQs) highlighted in [2]:

RQ1: what are the most common application domains for Big Data analytics, and how have they evolved over time?

RQ2: what are the major challenges and limitations that researchers have encountered in Big Data analysis, and how have they been addressed?

RQ3: what are the emerging research trends and directions in Big Data that will likely shape the field in the next 5 to 10 years?

In [2], we followed the methodology termed Systematic Literature Review (SLR), as proposed by Kitchenham and Charters in the field of software engineering [30].

Here, instead, we prompt Consensus with the three research questions, keep track and examine the bibliography detected, and evaluate the outputs and findings listed by Consensus to understand whether outputs highlighted by the manual SLR are similar to those detected by Consensus.

This practical experience also gave us the possibility to understand the pros and cons of adopting Consensus in conducting SLRs and how LLMs and GenAI can complement each other to support researchers in surveying the scientific literature in a rigorous way.

The paper is structured as follows: Section II reports the content generated by Consensus when prompted with the 3 SLR Research Questions. Section III compares the results obtained by the human-led SLR with the ones coming from Consensus. Section IV highlights the pros and cons of using Consensus for an SLR. Section V briefly introduces a framework to complement human-led SLRs with the support of GenAI tools. Section VI concludes the paper by summarizing the main contribution of the work.

II. ASKING GPT-4 CONSENSUS

To compare the power of Consensus in conducting a literature review, we started by prompting it with the 3 RQs listed in [2], and we kept track of the considerations reported by Consensus per each RQ as depicted in Figure 1. It was decided not to iteratively modify the prompt submitted to Consensus in order to avoid potential biases and suggestions

from domain experts (i.e., the authors of [2].) This leads to a clearer comparison between the human result and the AI-based result.

Hereafter, we report the output of each prompted question.

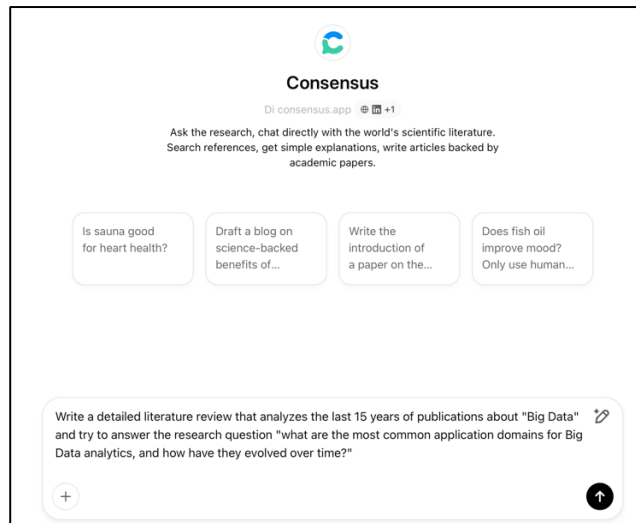


FIGURE 1. Consensus interface and prompting example.

A. RESEARCH QUESTION 1

The following prompt has been submitted to Consensus about Research Question 1 (RQ1):

Prompt for Question 1. *Write a detailed literature review that analyzes the last 15 years of publications about “Big Data” and try to answer the research question “what are the most common application domains for Big Data analytics, and how have they evolved over time?”*

Hereafter, the output generated by Consensus to answer RQ1:

1) INTRODUCTION TO RQ1

Big Data analytics has become a cornerstone of modern data-driven decision-making across numerous domains. Over the past 15 years, the application domains for Big Data analytics have evolved significantly, reflecting advancements in technology, increasing data availability, and changing business needs. This literature review aims to identify Big Data analytics' most common application domains and trace their evolution over time.

a: COMMON APPLICATION DOMAINS FOR BIG DATA ANALYTICS

1. Healthcare

- Applications: Big Data analytics in healthcare is used for predictive modeling, patient monitoring, and personalized medicine. It helps identify disease patterns, manage patient records, and improve clinical outcomes [8]
- Evolution: Initially focused on electronic health records (EHRs) and clinical data, the domain has expanded to

include genomics, real-time health monitoring through wearable devices, and telemedicine [9].

2. Finance

- Applications: In the financial sector, Big Data analytics is utilized for fraud detection, risk management, customer analytics, and algorithmic trading [10]
- Evolution: The focus has shifted from traditional data warehouses to real-time analytics and predictive modeling, driven by the need for rapid decision-making and enhanced security measures [11]

3. Retail and E-commerce

- Applications: Big Data analytics in retail and e-commerce includes customer segmentation, inventory management, personalized marketing, and demand forecasting [8]
- Evolution: Initially focused on historical sales data, the domain has incorporated real-time analytics, recommendation engines, and dynamic pricing strategies to enhance customer experience and operational efficiency [12].

4. Government and Public Sector

- Applications: Governments use Big Data analytics for policy-making, public health surveillance, crime prevention, and smart city initiatives [13]
- Evolution: The scope has widened from census data analysis and demographic studies to real-time data integration from IoT devices, social media analytics, and disaster response systems [14].

5. Manufacturing and Industry

- Applications: Big Data analytics in manufacturing includes predictive maintenance, supply chain optimization, and quality control [9]
- Evolution: From early uses in logistics and production data analysis, the applications have expanded to include advanced IoT integration, real-time monitoring, and smart manufacturing solutions [15]

2) EVOLUTION OVER TIME

a: EARLY YEARS (2009-2014)

The initial focus was on establishing the infrastructure for Big Data analytics, with significant investments in data storage and processing capabilities. Key technologies like Hadoop and MapReduce dominated the landscape, enabling batch processing of large datasets [16].

b: MIDDLE YEARS (2015-2018)

The period saw a shift towards real-time analytics and the integration of machine learning and deep learning techniques to extract more nuanced insights from data. There was a significant rise in domain-specific applications, particularly in healthcare, finance, and retail [8].

c: RECENT YEARS (2019-2023)

The focus has increasingly been on leveraging artificial intelligence (AI) and IoT for more sophisticated applications. There is a growing emphasis on predictive and prescriptive

analytics, driven by AI and machine learning advancements. Additionally, issues of data privacy and security have become more prominent as data volumes continue to grow [9].

3) CONCLUSION FOR RQ1

The application domains for Big Data analytics have expanded and evolved significantly over the past 15 years. Key domains include healthcare, finance, retail, government, and manufacturing, with each domain experiencing unique advancements and increasing integration of AI and real-time analytics. The evolution reflects a broader trend towards more sophisticated, domain-specific applications and a growing focus on real-time data processing and AI-driven insights.

B. RESEARCH QUESTION 2

The following prompt has been submitted to Consensus about Research Question 2 (RQ2):

Prompt for Question 2. *Write a detailed literature review that analyzes the last 15 years of publications about "Big Data" and try to answer the research question "what are the major challenges and limitations that researchers have encountered in Big Data analysis, and how have they been addressed?"*

Hereafter, the output generated by Consensus to answer RQ2:

1) INTRODUCTION TO RQ2

Big Data analytics has emerged as a critical component of modern data-driven decision-making processes. However, the immense volume, velocity, variety, and veracity of Big Data present numerous challenges and limitations. This literature review aims to identify the major challenges and limitations encountered in Big Data analysis over the last 15 years and discuss how researchers have addressed these issues.

a: MAJOR CHALLENGES AND LIMITATIONS IN BIG DATA ANALYSIS

1. Scalability and Storage Bottlenecks
 - Challenges: The massive size of Big Data requires scalable storage solutions and efficient processing capabilities. Traditional data storage systems often fall short in handling the volume and complexity of Big Data
 - Solutions: The development of distributed computing frameworks like Hadoop and Spark has been instrumental in addressing these scalability issues. These frameworks allow data to be processed in parallel across multiple nodes, significantly enhancing storage and processing capabilities [17].
2. Data Heterogeneity
 - Challenges: Big Data comes from diverse sources and in various formats, including structured, semi-structured, and unstructured data. This heterogeneity complicates data integration and analysis
 - Solutions: Techniques such as data lakes, virtual integration, and polystore architectures have been proposed to manage heterogeneous data. These solutions facilitate

the storage and querying of diverse data types within a unified framework [18].

3. Timeliness
 - Challenges: The high velocity of data generation necessitates real-time or near-real-time processing and analysis. Delays in data processing can render insights obsolete
 - Solutions: Stream processing platforms like Apache Kafka and real-time data processing tools like Apache Storm and Flink have been developed to address the need for timely data processing. These tools enable continuous data ingestion and real-time analytics [19].
4. Data Quality and Noise
 - Challenges: Big Data often contains significant amounts of noise, missing values, and errors, which can adversely affect the quality of analysis and the reliability of insights
 - Solutions: Data preprocessing techniques such as data cleaning, normalization, and imputation are critical for improving data quality. Machine learning algorithms have also been employed to identify and mitigate noise and errors in datasets [20].
5. Privacy and Security Concerns
 - Challenges: The aggregation of large volumes of data raises significant privacy and security issues. Ensuring data protection and compliance with regulations such as GDPR is a major concern
 - Solutions: Techniques such as data anonymization, encryption, and secure multi-party computation have been developed to enhance data privacy and security. Additionally, privacy-preserving data mining methods are being researched to enable safe data analysis without compromising individual privacy [21].
6. Algorithmic Complexity
 - Challenges: The complexity of algorithms required to analyze Big Data can be overwhelming. High-dimensional data and the need for sophisticated models make the development and implementation of algorithms challenging
 - Solutions: Advanced machine learning techniques, including deep learning, have been applied to handle the complexity of Big Data. These techniques can extract high-level features from large datasets, making it possible to uncover complex patterns and relationships [8].

2) ADDRESSING CHALLENGES OVER TIME

a: EARLY YEARS (2009-2013)

Focus was on establishing foundational infrastructure for Big Data analytics, with significant advancements in distributed computing and storage technologies. Techniques such as MapReduce were widely adopted to handle large-scale data processing [17].

b: MIDDLE YEARS (2014-2017)

Introduction of more sophisticated tools and frameworks to address the growing complexity and scale of Big Data.

Real-time processing tools and advanced machine-learning techniques gained prominence during this period [10].

c: RECENT YEARS (2018-2023)

Emphasis on integrating AI and machine learning with Big Data analytics to enhance predictive and prescriptive capabilities. Increased focus on privacy and security due to regulatory requirements and public concerns [19].

3) CONCLUSION FOR RQ2

The challenges and limitations of Big Data analysis are multifaceted, involving issues of scalability, data quality, privacy, and algorithmic complexity. Over the past 15 years, researchers have made significant strides in addressing these challenges through the development of new frameworks, algorithms, and technologies. Continuous advancements in machine learning and AI, along with improved data management practices, promise to mitigate these challenges further and unlock the full potential of Big Data analytics.

C. RESEARCH QUESTION 3

The following prompt has been submitted to Consensus about Research Question 3 (RQ3):

Prompt for Question 3. *Write a detailed literature review that analyzes the last 15 years of publications about "Big Data" and try to answer the research question "what are the emerging research trends and directions in Big Data that will likely shape the field in the next 5 to 10 years?"*

Hereafter, the output generated by Consensus to answer RQ3:

1) INTRODUCTION TO RQ3

Big Data has become a critical field of research, influencing various sectors with its capability to analyze vast amounts of data and provide actionable insights. Over the past 15 years, significant advancements have been made, and new research trends and directions have emerged. This literature review aims to identify and analyze these emerging trends and directions in Big Data that will likely shape the field in the next 5 to 10 years.

2) EMERGING RESEARCH TRENDS

1. Analytics-as-a-Service (AaaS)

- Trend: The concept of providing analytics capabilities over the cloud as a service has gained traction. AaaS allows organizations to leverage advanced analytics without substantial upfront investment in infrastructure
- Development: AaaS models are expected to evolve, offering more sophisticated analytics tools and capabilities, thus democratizing access to Big Data analytics [22].

2. Integration of AI and Machine Learning

- Trend: The integration of artificial intelligence (AI) and machine learning (ML) with Big Data analytics is one of the most significant emerging trends. These

technologies enable the development of predictive models and advanced data-processing techniques

- Development: AI and ML will continue to advance, with more emphasis on deep learning and reinforcement learning, providing more accurate predictions and insights from Big Data [8].
- #### 3. Real-Time Data Processing
- Trend: The need for real-time data processing is growing due to the increasing velocity of data generation from sources such as IoT devices and social media
 - Development: Future research will focus on improving real-time analytics frameworks to handle large-scale data streams efficiently, ensuring timely and actionable insights [23].
- #### 4. Blockchain Integration
- Trend: The integration of blockchain technology with Big Data analytics is an emerging trend that offers enhanced security and transparency in data transactions
 - Development: This integration will likely lead to the development of decentralized data storage and processing systems, improving data integrity and reducing the risk of fraud [24].
- #### 5. Edge Computing
- Trend: Edge computing involves processing data closer to its source rather than relying solely on centralized cloud servers. This approach reduces latency and bandwidth usage
 - Development: Research will focus on optimizing edge computing architectures for Big Data analytics, enabling real-time processing and decision-making at the data source [25].
- #### 6. Data Privacy and Security
- Trend: With the increasing volume of data, ensuring data privacy and security is becoming more challenging and critical
 - Development: Future research will likely focus on developing advanced encryption techniques, privacy-preserving data mining methods, and robust data governance frameworks to protect sensitive information [26].
- #### 7. Interdisciplinary Applications
- Trend: Big Data analytics is increasingly being applied across various disciplines such as healthcare, finance, education, and social sciences
 - Development: Interdisciplinary research will continue to grow, with more customized analytics solutions being developed to address specific challenges in different fields [16].
- #### 8. Predictive and Prescriptive Analytics
- Trend: There is a growing emphasis on not just understanding past data (descriptive analytics) but also predicting future trends (predictive analytics) and recommending actions (prescriptive analytics)
 - Development: The development of more advanced algorithms and models will enhance the capabilities of predictive and prescriptive analytics, making them more accurate and actionable [27].

9. Visualization and User Interfaces
- Trend: Effective visualization tools are essential for making Big Data insights accessible and understandable to non-technical users
 - Development: Future research will focus on developing more intuitive and interactive visualization tools that can handle the complexity and scale of Big Data [28].
10. Sustainability and Green Computing
- Trend: The environmental impact of Big Data processing is becoming a concern, with a push towards more energy-efficient computing practices
 - Development: The research will explore sustainable computing practices, focusing on reducing the carbon footprint of data centers and developing energy-efficient algorithms and hardware [29].

3) CONCLUSION FOR RQ3

The field of Big Data is rapidly evolving, with numerous emerging trends and research directions poised to shape its future. The integration of AI and ML, real-time data processing, blockchain, edge computing, and a focus on data privacy and security are some of the key trends that will drive advancements in the next 5 to 10 years. As these trends develop, they will not only enhance the capabilities of Big Data analytics but also broaden its applications across various domains, ensuring that Big Data continues to be a vital tool for innovation and decision-making.

III. COMPARING GPT-4 CONSENSUS VS. MANUAL SLR

Our manual SLR detected 207 related works, 32 of which were identified as Primary Studies (PSs) related to the 3 RQs. Consensus detected 22 PSs (from 2011 to 2023) to address all 3 RQs (9 PSs for RQ1, 5 PSs for RQ2, and 8 PSs for RQ3). After a careful manual check (i.e., we applied the inclusion/exclusion criteria highlighted in [2] to verify all those primary studies that do not satisfy the exclusion criteria above and thus are to be considered as not valid PSs), one publication discovered as PS by Consensus is not available on the web ([14]), resulting in a clear example of hallucination. Eight publications were listed by Consensus with errors on the list of Authors ([10], [11], [12], [13], [15], [17], [18], [19]), while seven publications contain errors on the year of publication ([11], [13], [18], [19], [20], [21], [27]). Five of the listed PSs are not available as primary documents in Scopus. Section “Consensus References” reports only the corrected references and publication years after our manual check. Six Consensus PSs are in common with our SLR, while the remaining 15 valid Consensus PSs were not identified and discussed in our SLR. Contrariwise, 201 related works were not discussed by Consensus. None of the 32 PSs detected by the SLR were taken into consideration by Consensus.

Our manual SLR’s PSs [2] generated 2062 total citations (max citations: 317 [26]). The 21 Consensus’s PSs generated 4444 total citations (max citations: 1750 [8]).

Figure 5 in [2] and Figure 2 shows that the number of SLR PSs and citations are mainly condensed in 2020 and 2021, while for Consensus, we have the following publication year distribution: 1 paper in 2011, 2012, 2013, 2 papers in 2015, 2016, and 2017, 3 papers in 2018, 1 paper in 2019, 3 papers in 2020 and 2021, and 1 paper in 2022 and 2023. As depicted in Figure 2, PSs selected by Consensus were mainly published in the year 2020 and 2021.

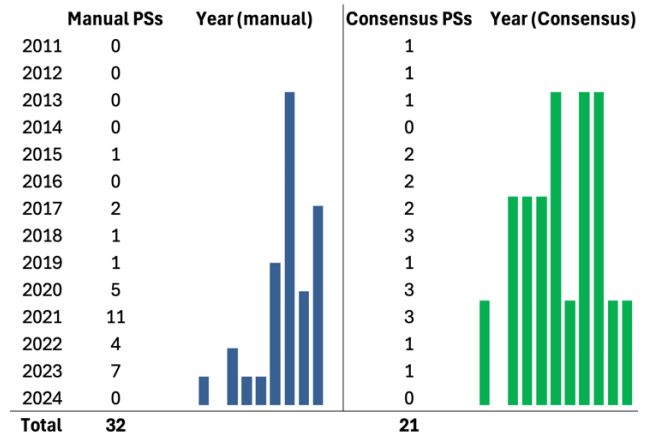


FIGURE 2. Year distribution for PSs detected manually and also by Consensus.

As for RQ1, the SLR identified and discussed 7 application domains: Business and Finance; Healthcare; Retail; Manufacturing; Telecommunications; Government, Public Services, and Transportation; Media, Entertainment, and Education. Consensus identified and discussed 5 application domains without covering the impact of Big Data in the TLC sector, Media, Entertainment, and Education sectors.

As for RQ2, the SLR identified and discussed 4 main research challenges: the integration with IoT systems; the management of Security and Privacy issues; the focus on social media and data mining aspects; and the importance of data quality and scalability. Consensus focused on 6 challenges about infrastructures able to manage big data, data heterogeneity, and their quality; and the decrease of algorithmic complexity.

As for RQ3, the SLR identified and discussed 6 emerging research trends: the integration of new and advanced technologies; the evolution and specialization of new application domains; the need for real-time analytics; the adoption of AI and cognitive computing; the advancement in data security and privacy; and the exploration of multimodality. Consensus focused on 10 trends related to the development of platforms for offering analytics as a service; the integration of AI and ML facilities for predictive and prescriptive analysis; the integration of blockchain and edge computing technologies; the advancement in data security and privacy; and a clear focus on green computing.

More rigorous and objective metrics have been introduced to assess the effectiveness of AI-generated literature reviews compared to human-led SLRs to strengthen the quantitative

analysis. Precision and Recall for Study Detection have been adopted. We compared the number of AI-identified PSs to those found in our human-led SLR. Precision, recall, and F1-score based on the overlap and differences between the AI-selected and manually selected studies have been computed:

Precision (Positive Predictive Value): Measures how many of the studies identified by Consensus as relevant (Primary Studies, PSs) are correct when compared to the manually selected PSs.

Precision

$$= \frac{\text{True Positives TP (Correctly identified PSs)}}{\text{TP} + \text{False Positives FP (AI-selected PSs not in SLR)}}$$

A high precision means that most of the AI-selected studies are indeed relevant. A low precision indicates that the AI is including many irrelevant studies (false positives).

Recall (Sensitivity): Measures how many of the actual relevant studies (PSs manually identified) were successfully detected by Consensus.

$$\text{Recall} = \frac{\text{TP (Correctly identified PSs)}}{\text{TP} + \text{False Negatives FN (Missed PSs)}}$$

A high recall means the AI is capturing most of the relevant studies. A low recall means the AI is missing a significant portion of important studies.

F1-Score: The harmonic mean of precision and recall, providing a single metric to evaluate AI performance.

$$F1 - \text{score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

A balanced F1 score indicates that the AI is both accurate and comprehensive.

The key quantitative findings and metrics are listed in Table.

TABLE 1. Precision, Recall, and F1-score of consensus.

	Manual SLR	Consensus AI
Valid PSs	32	21
Overlapping TP	6	6
Other TP	32-6 = 26	21-6-5 = 10
False Positive	0	5
False Negative	10	26
Precision	100%	76%
Recall	76%	38%
F1-score	86%	51%

IV. PROS AND CONS IN USING GPT-4 CONSENSUS

This comparison (see Table 2) clearly highlights that LLM and GenAI solutions can effectively complement and enhance manual SLRs. Based on our practical experience, here we outline a set of key advantages (pros) and limitations (cons) of using Consensus as follows:

To make the limited critical appraisal of Consensus more tangible, some examples are listed (Table 3) to clearly demonstrate why human critical analysis remains essential in SLRs and why AI, while useful, should be integrated with human oversight.

TABLE 2. Advantages (pros) and limitations (cons) of using consensus.

Pros of Using Consensus	Cons of Using Consensus
Speed and Efficiency: Consensus can quickly retrieve and summarize relevant research papers, saving significant time compared to the lengthy process of a traditional SLR	Lack of Depth: While Consensus provides a broad overview, it may not delve as deeply into the methodologies and details of each study compared to an SLR
Accessibility: The platform is user-friendly and accessible, making it easier for researchers without extensive experience in systematic reviews to gather and synthesize relevant literature	Quality Control: SLRs include rigorous quality assessments of selected Primary Studies, which might be less stringent or absent in automated tools like Consensus
Real-Time Updates: Consensus provides access to the most recent studies and findings, ensuring that the literature review includes up-to-date information. This is also true for SLRs conducted over well-known scientific DBs	Potential Bias: Automated tools may introduce biases based on their algorithms and the databases they access, which might not be as thoroughly vetted as manually curated systematic reviews
Comprehensive Coverage: With the ability to search across multiple databases and sources, Consensus can offer a broad view of available literature in line with well-conducted SLRs	Limited Critical Appraisal: Consensus may lack the critical appraisal aspect that is a cornerstone of systematic reviews, potentially overlooking study limitations and biases
	Hallucinations: Consensus may produce hallucinations during literature review (i.e., the model generates information that is incorrect, and sometimes it invents bibliographic sources), thus still making it important to deeply cross-check manually the findings and the studies selected.
Free access to scientific literature. For manual SLR, accessing scientific databases (such as Scopus, IEEE, ACM) requires institutional subscriptions.	
Reduced search and research time if compared to traditional SLRs that require 3 person months in general.	

TABLE 3. Example of AI limitations in critical analysis.

Limitation	Example
Hallucinated or incorrect references	Reference [14] was non-existent; 5 other PSs had incorrect metadata
Superficial challenge identification	Missed IoT integration as a challenge in RQ2
Gaps in application domain coverage	Missed Telecommunications, Media & Education in RQ1
Bias toward citation count	AI-selected PSs had 3x higher avg. citations but ignored all 32 human-selected PSs
Failure to compare conflicting studies	Presented blockchain as a future trend without addressing scalability challenges

V. HINTS IN COMPLEMENTING MANUAL SLR WITH GENAI SYSTEMS

Hereafter, we suggest a possible integration of manual SLRs enriched with GenAI support. Figure 3 summarizes the manual steps for SLRs and highlights the sub-steps where GenAI can complement the manual process of surfing and analyzing literature. Researchers can streamline specific phases of

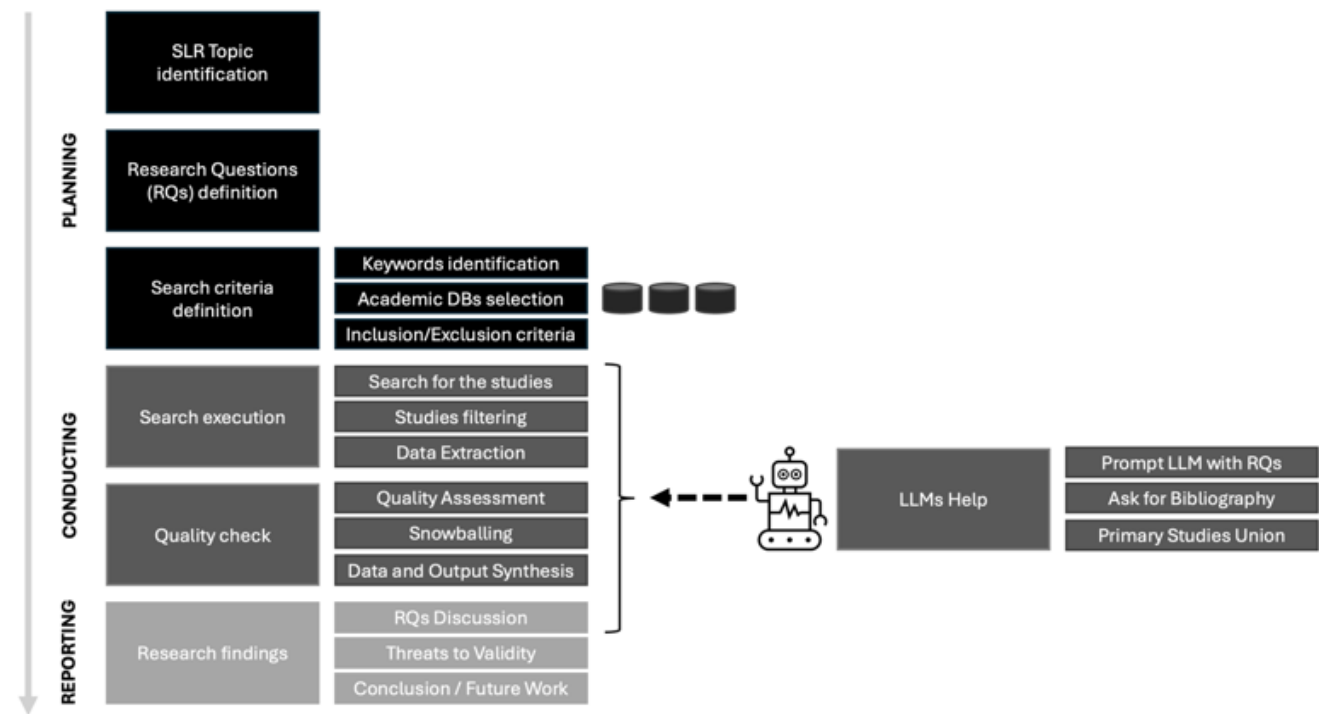


FIGURE 3. SLR’s execution steps enriched with Gen-AI and LLMs support.

the SLR process, particularly the literature search and study selection stages. This integration, however, must be carried out thoughtfully, ensuring that the benefits of GenAI’s speed and scale are balanced against the need for human critical appraisal and manual checks to avoid biases, inaccuracies, and errors. As shown in Figure 3, the traditional SLR process can be broken down into three main phases:

Phase1 - Planning the SLR: This phase involves defining the research objectives, identifying relevant research questions, and determining the keywords to be used in the literature search.

Phase2 - Conducting the SLR: Researchers surf databases and research libraries, filter relevant studies, apply inclusion/exclusion criteria, and assess the quality of studies.

Phase3 - Reporting the Results: In the final phase, findings are synthesized and reported. Researchers discuss answers to the research questions, and potential threats to validity are addressed.

While the planning and reporting phases largely remain manual processes, GenAI can play a significant role in assisting researchers during Phase2 “Conducting the SLR”. Specifically, GenAI can support researchers in the following areas:

1. Literature Search and Study Detection: During the literature search stage, the LLM can be prompted using the keywords and research questions established in the planning phase. The model can then automatically retrieve and present a list of relevant studies. This dramatically reduces the time spent manually searching

databases and ensures that a broader range of potential sources is considered beyond the scope of manual searches. However, researchers should manually verify the AI-generated bibliography to avoid “hallucinations,” where the model might present incorrect or invented references.

2. Filtering and Quality Assessment of Studies: Researchers can filter these results manually by applying inclusion and exclusion criteria once the LLM provides a bibliography. This step is crucial for ensuring the quality of the selected studies and avoiding irrelevant or low-quality literature. Although LLMs can be used to summarize the content of papers, they still require human oversight to assess methodological soundness, the credibility of sources, and relevance to the SLR topic. This manual quality check is essential in preventing the introduction of biases or errors that might occur if LLM outputs are accepted without critical review.
3. Combining GenAI Results with Manual Detection: After the LLM-assisted literature search and filtering process, the remaining studies can be cross-referenced with the ones manually retrieved by researchers. This hybrid approach ensures that no relevant study is overlooked while still leveraging the comprehensive search capabilities of GenAI. It also minimizes the risk of excluding studies that an LLM might have missed or misinterpreted due to limitations in its training data.

Although LLMs can provide a broad, rapid synthesis of existing literature, one of their main limitations lies in their

ability to appraise the quality of research studies critically. This task, which requires a deep understanding of research methodologies and domain-specific knowledge, is still best handled by human experts. Researchers must, therefore, carefully study the LLM-identified Primary Studies to ensure that they meet the required standards of rigor and quality. Human intervention is particularly important to:

- Evaluating study design and methodology: LLMs may not be adept at assessing whether a study employs a robust experimental design or is prone to biases, such as selection or publication bias
- Interpreting nuanced findings: LLMs can summarize large amounts of data but may struggle with interpreting subtle or context-dependent conclusions. Researchers must ensure that the AI-generated summaries align with the true meaning of the findings
- Maintaining critical analysis: LLMs may prioritize quantity over depth, and without human oversight, there is a risk that critical insights could be missed.

In conclusion, a multi-step validation process should be integrated into the workflow to manually verify the sources provided by the LLM, checking the existence and validity of the cited studies in recognized databases. Moreover, researchers should critically assess whether the studies identified by the LLM meet the inclusion, exclusion, and quality criteria, such as relevance to the research questions and robustness of the methodology.

VI. CONCLUSION AND FUTURE RESEARCH

In summary, a comparison between the results of a manual systematic literature review versus GPT-4 Consensus results is discussed. Moreover, the pros and cons of adopting LLMs for SLRs are highlighted, and a framework to complement manual SLRs with the support of LLM systems is introduced.

By incorporating GenAI into the literature search and selection phases, the overall efficiency of SLRs can be significantly improved. GenAI models can reduce the time required to screen and retrieve studies from databases and may help uncover relevant research that manual searches might overlook. This increased speed allows researchers to focus more on tasks that require deeper critical thinking, such as synthesis, interpretation, and reporting.

The hybrid approach—combining GenAI assistance with human oversight—represents an optimal strategy for ensuring that SLRs are both comprehensive and accurate. Researchers can take advantage of the strengths of GenAI models, particularly their ability to process and summarize vast amounts of data quickly while still maintaining control over the quality and reliability of the literature review process.

While Consensus offers a faster and more accessible way to conduct literature reviews with real-time updates and broad coverage, it may fall short in terms of depth, quality control, and critical appraisal compared to traditional SLRs. Expert human intervention is needed to supervise the help coming from LLMs in making literature reviews.

Future research should investigate AI-assisted SLR performance in diverse academic fields to fully assess the generalizability of AI-powered literature reviews. Some additional key areas for further exploration include (but are not limited to):

1. Comparative Studies: Conducting side-by-side evaluations of AI-assisted SLRs in healthcare, finance, and social sciences to compare AI accuracy and biases.
2. Domain-Specific AI Models: Exploring whether specialized LLMs trained on field-specific datasets (e.g., biomedical AI models for medical research) improve accuracy over general-purpose models like GPT.
3. Comparative GenAI: Conducting ad-hoc comparisons among different GenAI engines.
4. Hybrid AI-Human Review Models: Developing frameworks where AI supports but does not replace traditional human-led SLR processes, ensuring reliability while leveraging AI's efficiency.
5. Bias and Ethical Considerations: Investigating how AI biases impact literature selection in different fields and developing strategies to mitigate these biases.

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The author declare that GenAI Consensus has been used to conduct the SLR. The Consensus's output is reported in Section II.

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